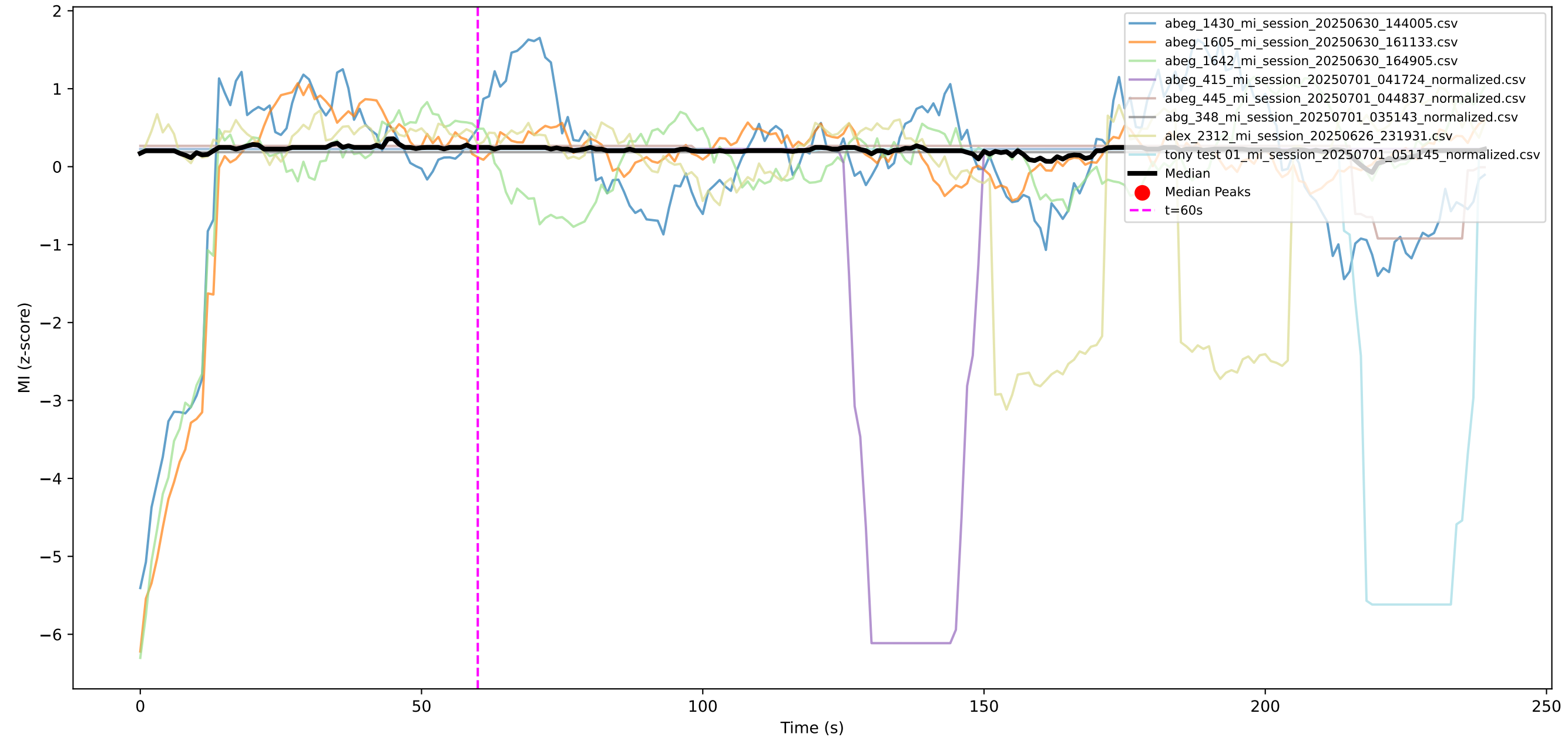
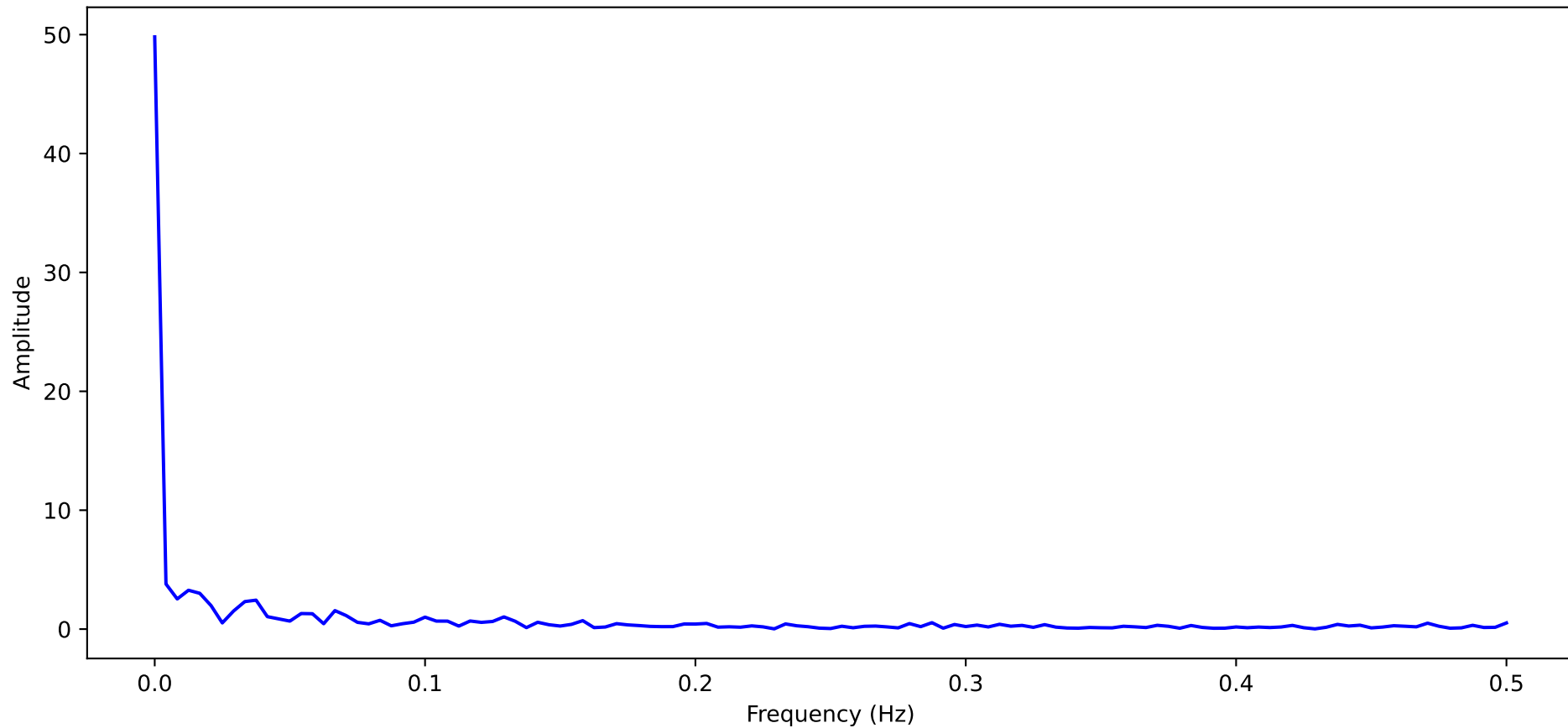


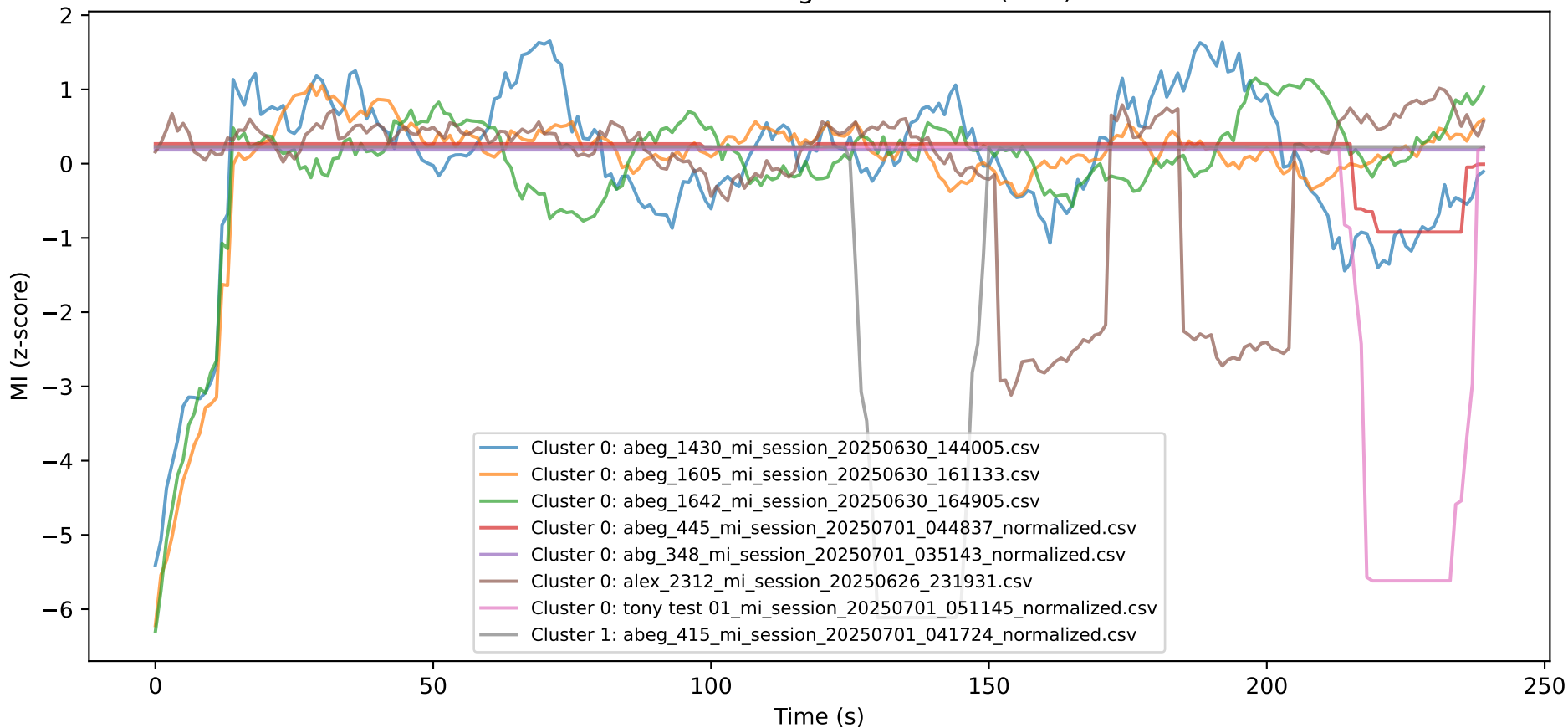
MI Curves (z-score, MA=20) with Median and Peaks



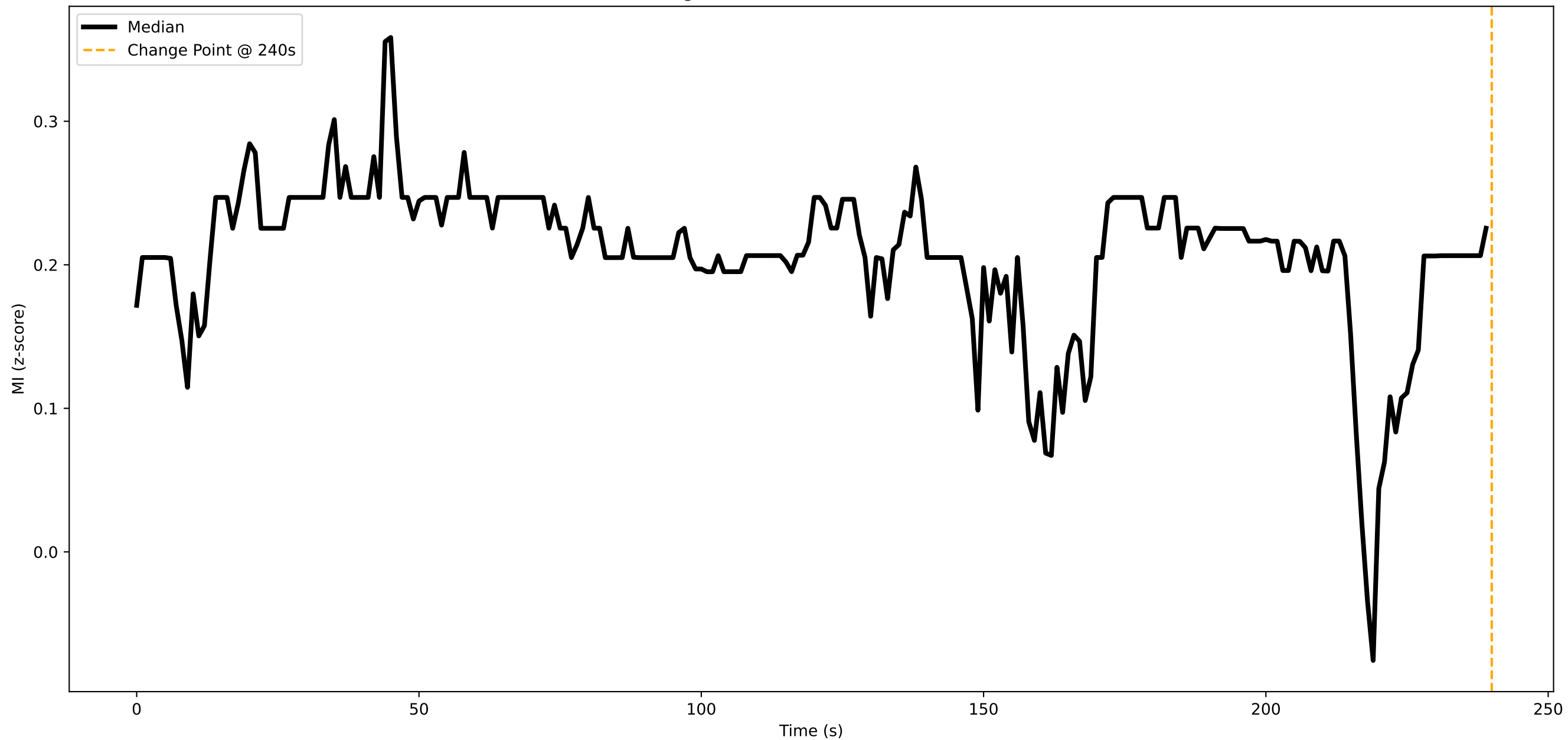
Median MI Frequency Spectrum (FFT)



KMeans Clustering of MI Curves (k=2)



Change Point Detection on Median MI



MI Advanced Report

Generated: 20250716_232112

Statistical Interpretation:

- All MI session files with ≥ 240 samples were included.
- Each series was smoothed (moving average window=20) and z-score normalized.
- Each file is plotted in a unique color, with the median curve in black.
- $t=60s$ is marked for reference.
- Peak analysis: Median curve has 0 peaks ($z > 1.5$) at times [].
- AUC (median): 49.61
- Bootstrap 95% CI for mean diff (first vs. second half): [-0.01319985 0.01387886]
- Change points detected at: [240]
- FFT shows frequency content of median MI.
- KMeans clustering ($k=2$) groups curves by similarity.
- Mixed-effects model:

Mixed Linear Model Regression Results

```
=====
Model:          MixedLM Dependent Variable: value
No. Observations: 1920  Method:          REML
No. Groups:      8      Scale:           1.2493
Min. group size: 240    Log-Likelihood: -2953.1571
Max. group size: 240    Converged:       Yes
Mean group size: 240.0
```

```
-----
          Coef. Std.Err.  z    P>|z| [0.025 0.975]
-----+-----
Intercept    0.077    0.077  0.997  0.319 -0.075  0.229
time        -0.001    0.000 -2.835  0.005 -0.002 -0.000
Group Var     0.027    0.016
```

- ```
=====
```
- ANOVA across files: ANOVA  $F=6.21$ ,  $p=0.0000$